PEARSON EDEXCEL INTERNATIONAL GCSE

# Mathematics A

## Unit 2 • Chapter OL-T55 Classified

**Cross-session • chapter-organised candidate**

22 classified items • 79 marks • 1 chapter(s)

**Source-faithful practice with synchronized solutions**

**Every question is followed by its worked answer/reference**


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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

# Contents

Unit 2 • Unit 2 • Chapter OL-T55 • 22 items • 79 marks



June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# Cumulative Frequency Diagrams

**Unit 2 • 79 marks • 22 item(s)**

**June 2025 | 4MA1-2H Q13 | 6 marks**
Plot a cumulative-frequency curve from class frequencies

**June 2024 | 4MA1-2H Q13 | 7 marks**
Estimate the number above or between thresholds

**November 2023 | 4MA1-1H Q13 | 1 marks**
Estimate the median from a cumulative-frequency diagram

**November 2023 | 4MA1-1H Q13 | 3 marks**
Estimate a percentage from a cumulative curve

**June 2023 | 4MA1-1H Q12 | 4 marks**
Estimate the median from a cumulative-frequency diagram

**November 2024 | 4MA1-2H Q13 | 6 marks**
Estimate the number above or between thresholds

**June 2022 | 4MA1-2H Q13 | 7 marks**
Estimate the number above or between thresholds

**January 2022 | 4MA1-1H Q12 | 6 marks**
Estimate a percentage from a cumulative curve

**January 2020 | 4MA1-1H Q12 | 2 marks**
Estimate IQR from a cumulative-frequency diagram

**January 2020 | 4MA1-1H Q12 | 3 marks**
Estimate a percentage from a cumulative curve

**January 2021 | 4MA1-1H Q11 | 2 marks**
Estimate the median from a cumulative-frequency diagram

**January 2021 | 4MA1-1H Q11 | 3 marks**
Recover a threshold from a target count or amount

**June 2021 | 4MA1-2H Q12 | 2 marks**
Plot from an existing cumulative-frequency table

**June 2021 | 4MA1-2H Q12 | 1 marks**
Estimate the median from a cumulative-frequency diagram

**June 2021 | 4MA1-2H Q12 | 2 marks**
Estimate IQR from a cumulative-frequency diagram

**... 7 more item(s) follow**

**Original question context and synchronized companion pages follow.**

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

**13** The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

| **Time ($t$ minutes)** | **Frequency** |
|---|---|
| $0 < t \leqslant 10$ | 8 |
| $10 < t \leqslant 20$ | 13 |
| $20 < t \leqslant 30$ | 12 |
| $30 < t \leqslant 40$ | 17 |
| $40 < t \leqslant 50$ | 7 |
| $50 < t \leqslant 60$ | 3 |

(a) Complete the cumulative frequency table.

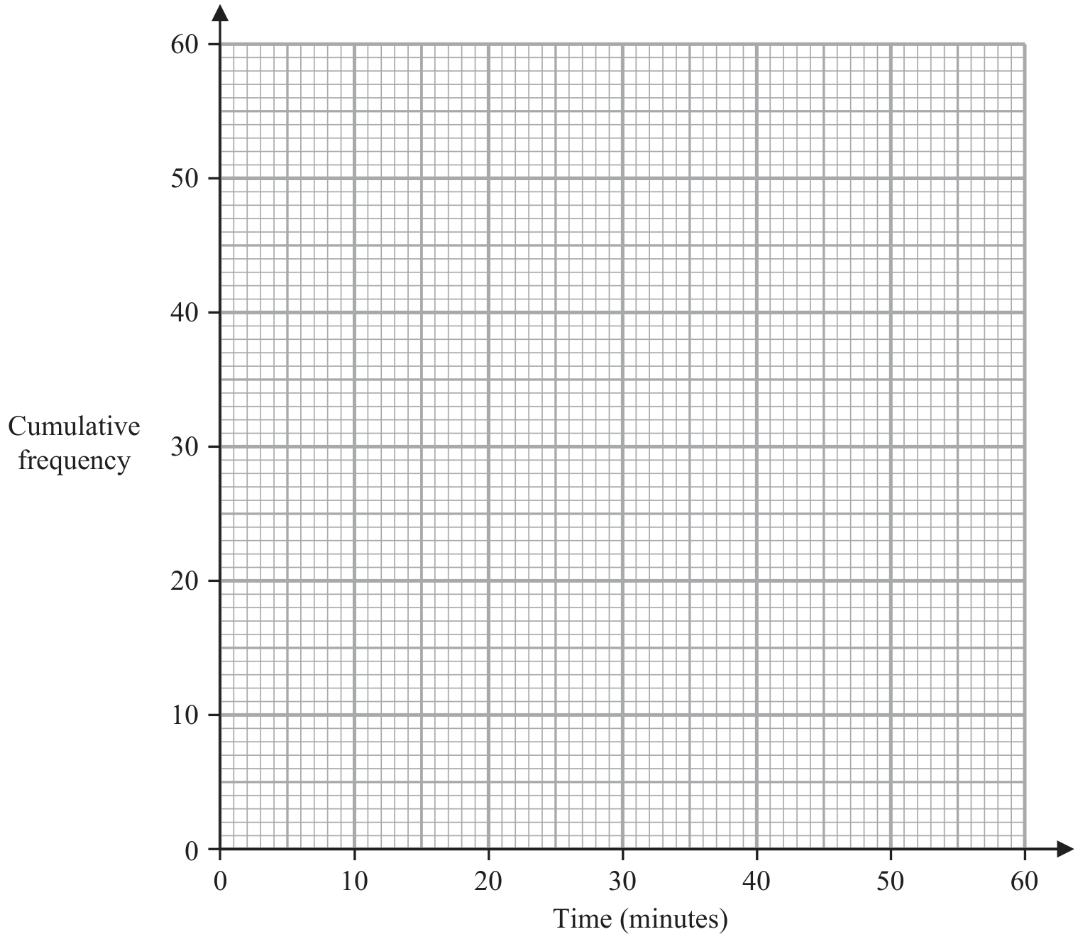
| **Time ($t$ minutes)** | **Cumulative frequency** |
|---|---|
| $0 < t \leqslant 10$ | |
| $0 < t \leqslant 20$ | |
| $0 < t \leqslant 30$ | |
| $0 < t \leqslant 40$ | |
| $0 < t \leqslant 50$ | |
| $0 < t \leqslant 60$ | |

**(1)**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

## Continued
4MA1-2H Q13

ORIGINAL QUESTION




**(2)**

(c) Use your graph to find an estimate for the median time.

.............................. minutes
**(1)**

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

..............................................................
**(2)**

**(Total for Question 13 is 6 marks)**

# Worked Solution 26

Cumulative frequency graph and estimates

UNIT 2
6 marks

## Cumulative totals

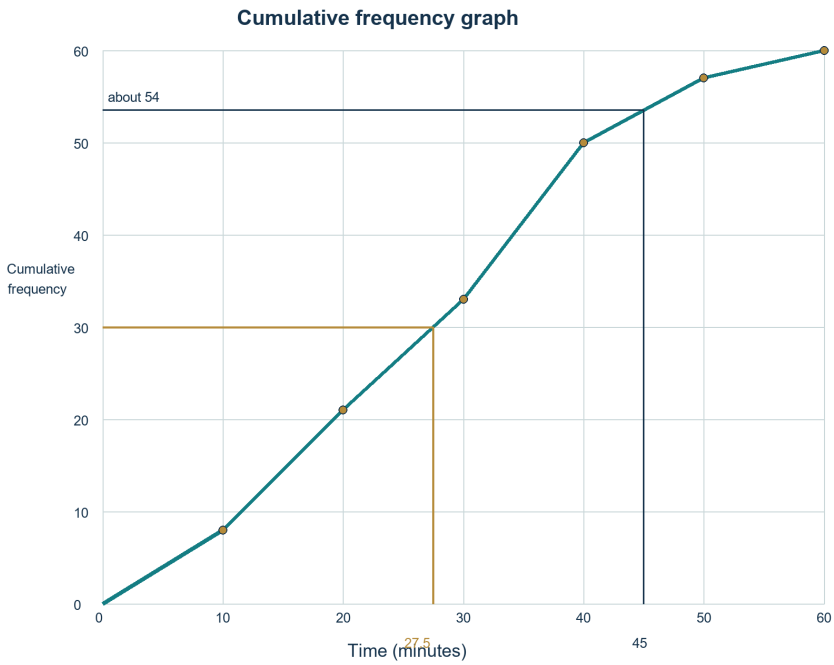
$$8,\ 8+13=21,\ 33,\ 50,\ 57,\ 60.$$

## Plot

$$(10,8),(20,21),(30,33),(40,50),(50,57),(60,60),\ \text{joined as a cumulative-frequency graph.}$$

## Read estimates

$$\text{Median: 27 to 28 minutes;}\qquad \text{more than 45 minutes: } 60-54\approx 6 \text{ people.}$$




## Final answer

$$8,21,33,50,57,60;\quad \text{median 27 to 28 min;}\quad \text{about 6 people}$$

## Question 18
4MA1-2H Q13

ORIGINAL QUESTION
**7 marks**

**13** The table gives information about the distances, in km, that 70 teachers travel to school.

| **Distance (*d* km)** | **Frequency** |
|---|---|
| $0 < d \leqslant 10$ | 7 |
| $10 < d \leqslant 20$ | 17 |
| $20 < d \leqslant 30$ | 18 |
| $30 < d \leqslant 40$ | 14 |
| $40 < d \leqslant 50$ | 10 |
| $50 < d \leqslant 60$ | 4 |

(a) Complete the cumulative frequency table.

| **Distance (*d* km)** | **Cumulative frequency** |
|---|---|
| $0 < d \leqslant 10$ | |
| $0 < d \leqslant 20$ | |
| $0 < d \leqslant 30$ | |
| $0 < d \leqslant 40$ | |
| $0 < d \leqslant 50$ | |
| $0 < d \leqslant 60$ | |

**(1)**

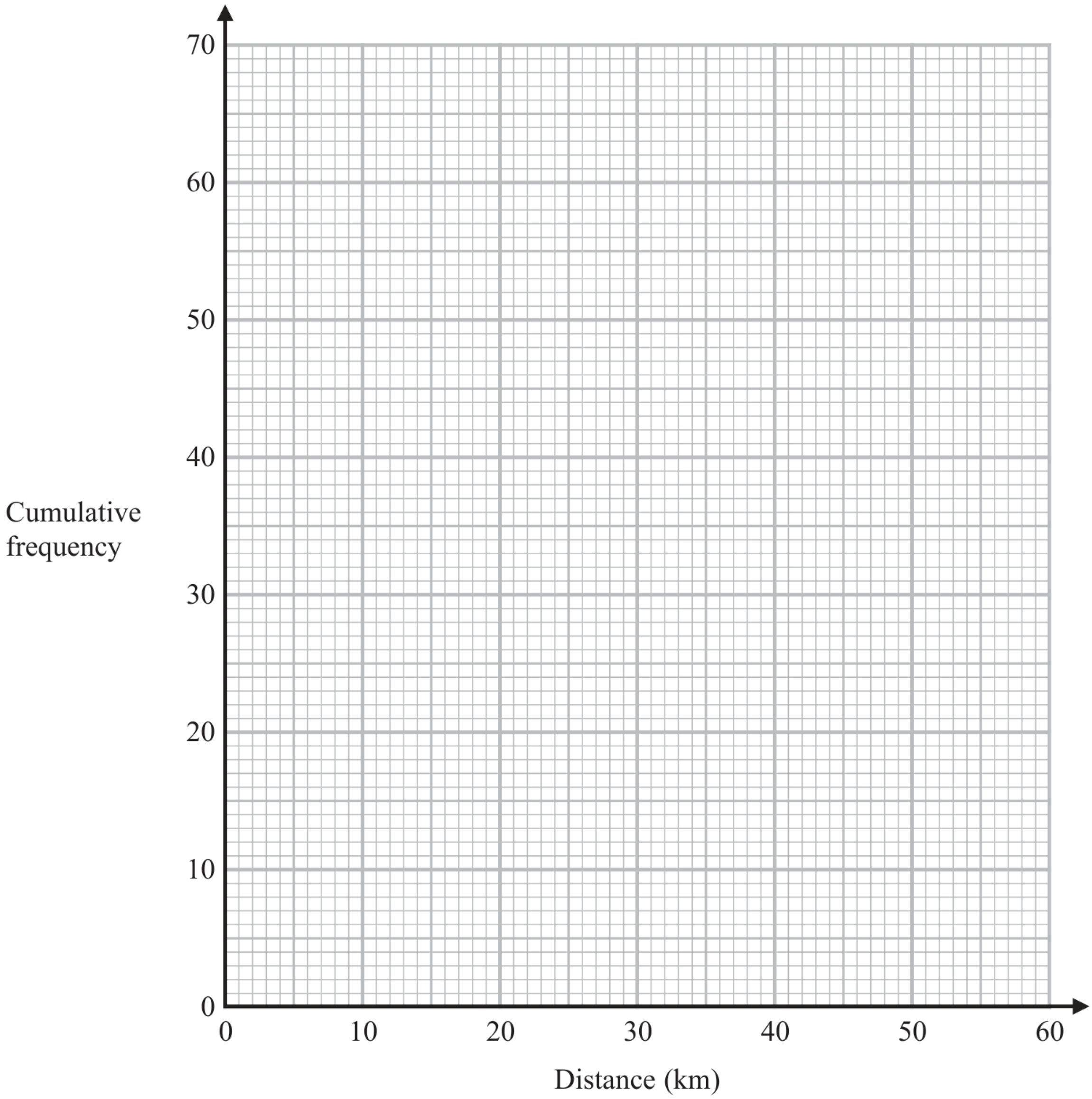
## Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION
**7 marks**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

**(2)**




(c) Use your graph to find an estimate for the interquartile range of the distances.

.................................................... km

**(2)**

## Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION
**7 marks**

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

........................................................

**(2)**

## Worked Solution 18

Cumulative frequency

MODULAR UNIT 2
**7 marks**

### Table

$$7, 24, 42, 56, 66, 70.$$

### Graph

Plot $(10, 7), (20, 24), (30, 42), (40, 56), (50, 66), (60, 70)$ and join smoothly.

### IQR

$$Q_1 \approx 16.2,\ Q_3 \approx 37.5 \Rightarrow \text{IQR} \approx 21.3 \text{ km}.$$

### More than 46 km

$$70 - \text{CF}(46) \approx 70 - 62 = 8.$$

### Final answer

CF $7, 24, 42, 56, 66, 70$; IQR $\approx 21.3$ km; $8$ teachers

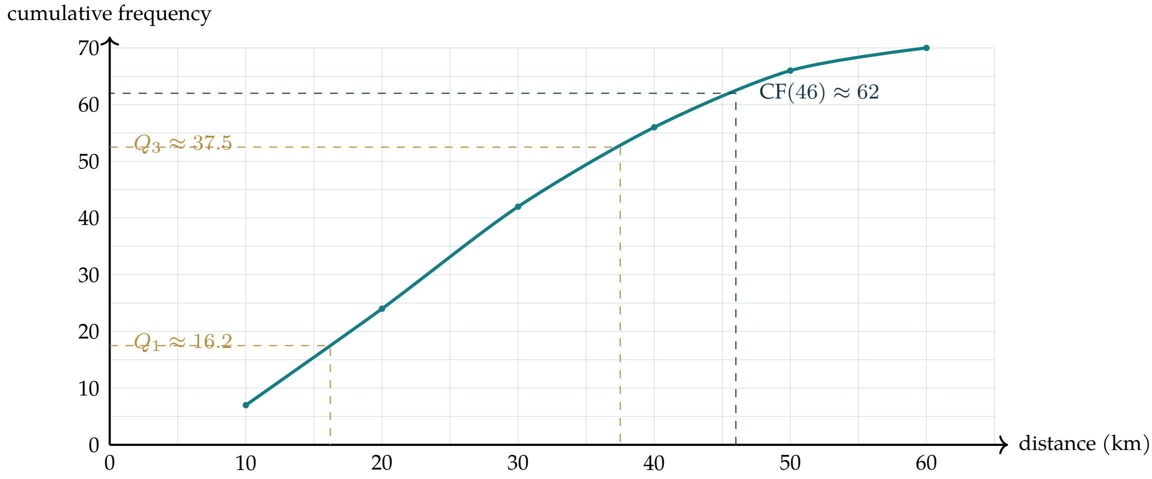
# Completed diagram 18

Cumulative frequency

MODULAR UNIT 2

7 marks

## Completed solution diagram: Cumulative-frequency graph



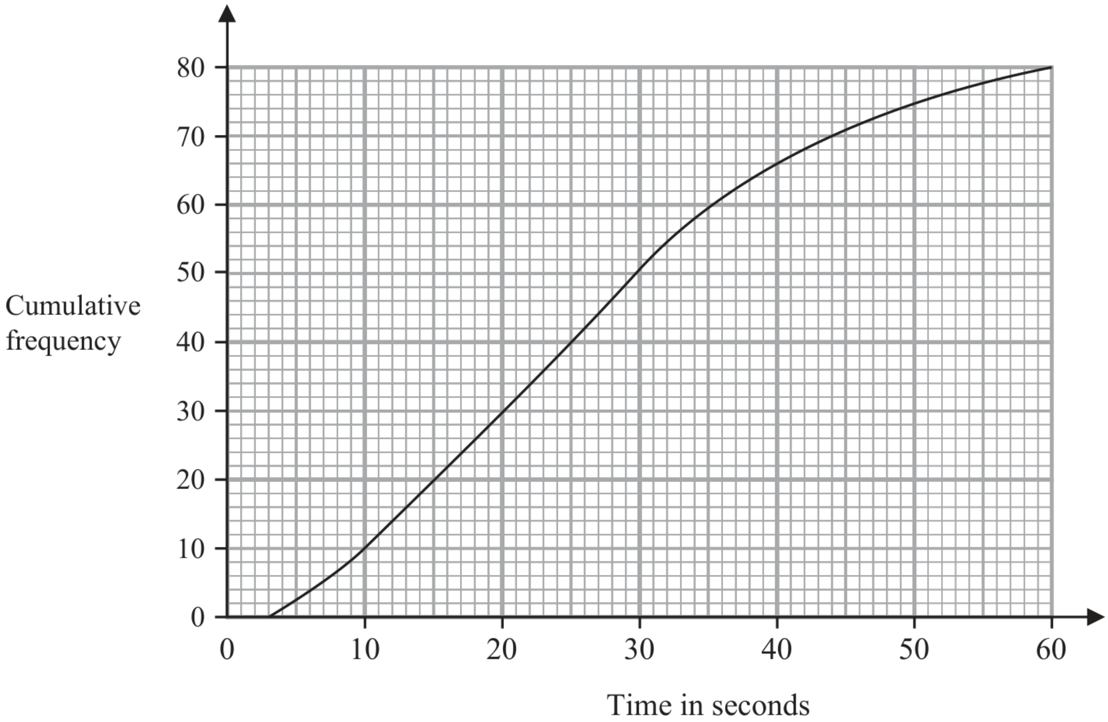
## Question 09

4MA1-1H Q13 (a)

ORIGINAL QUESTION UNIT 2

**1 marks**

**13** The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.




(a) Find an estimate for the median time.

.................................................. seconds

**(1)**

## Worked solution 09

Median from cumulative frequency

MODULAR UNIT 2
**1 marks**

### (a) – Median from cumulative frequency

**Read the middle rank**

$$80/2 = 40; readingCF = 40fromthegraphgivesabout25seconds.$$

**Final answer**

25 seconds

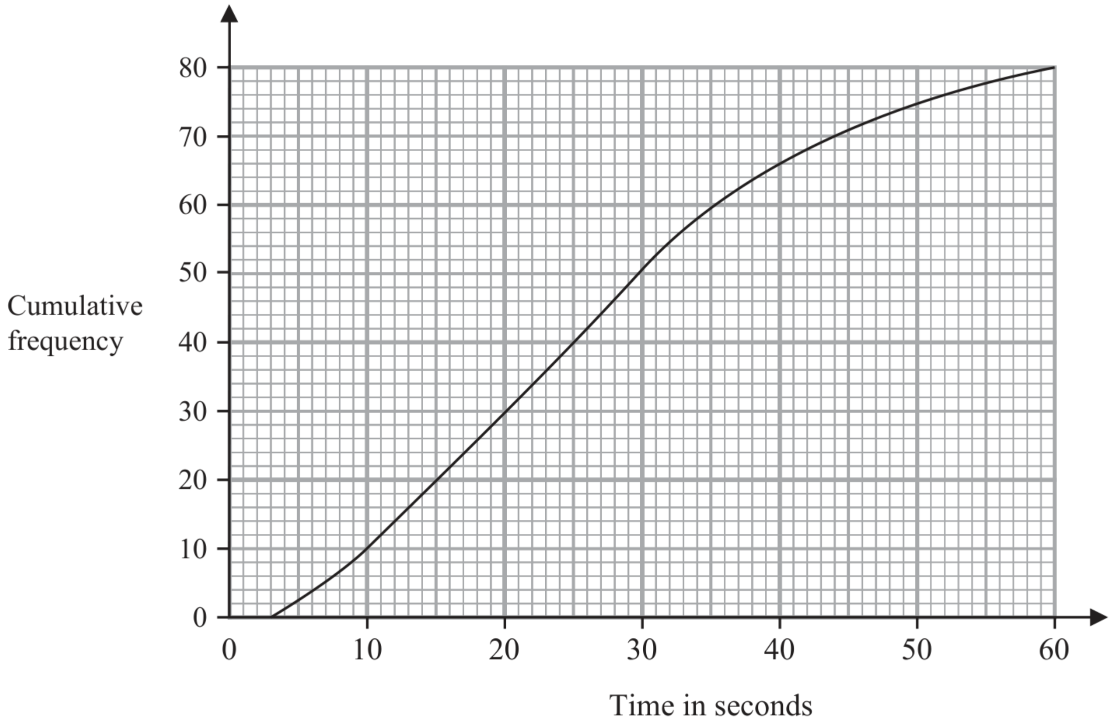
## Question 10

4MA1-1H Q13 (b)

ORIGINAL QUESTION UNIT 2

**3 marks**

**13** The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.




(b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

.......................................................%

**(3)**

## Worked solution 10

Percentage taking longer than 50 seconds

MODULAR UNIT 2
**3 marks**

### (b) – Percentage taking longer than 50 seconds

**Read CF at 50**

$$CF(50) \approx 75, \quad 80 - 75 = 5.$$

**Convert to a percentage**

$$\frac{5}{80} \times 100 = 6.25\%.$$

### Final answer

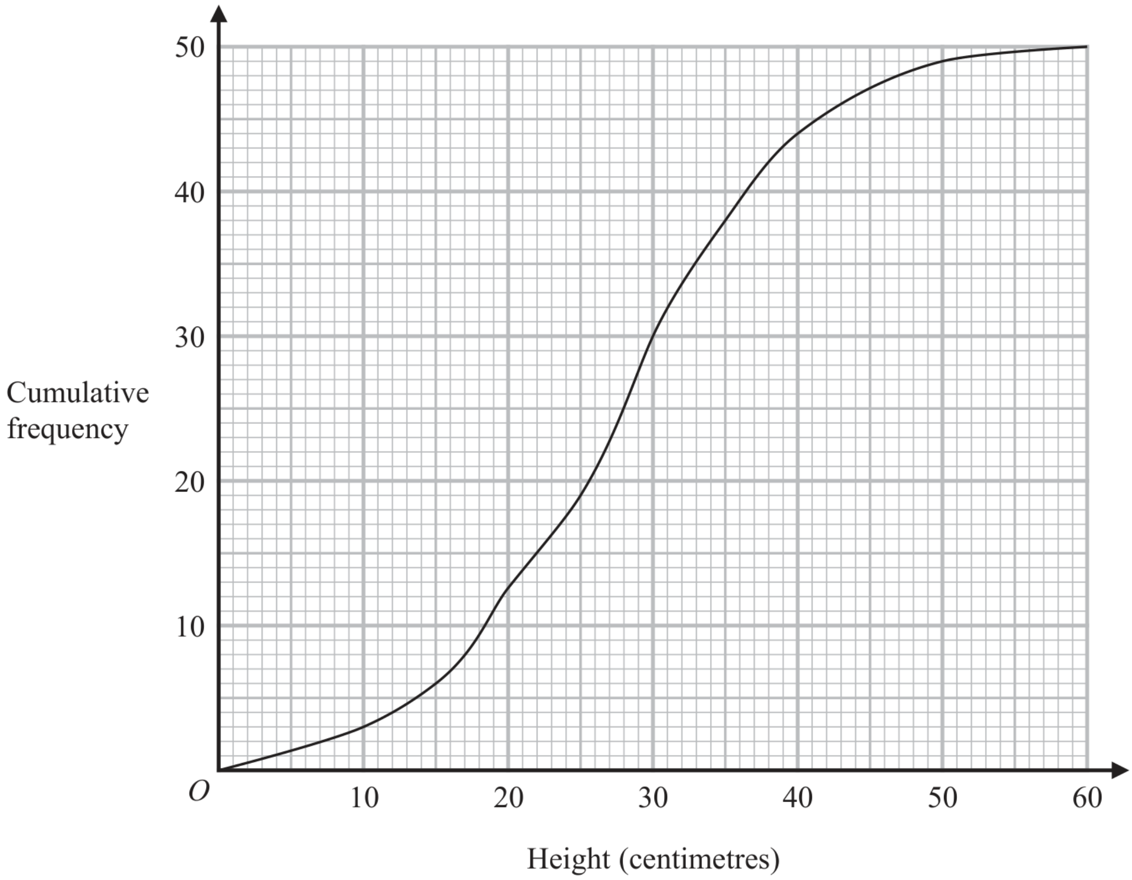
6.25%

## Question 12

4MA1-1H Q12

ORIGINAL QUESTION
**4 marks**

**12** The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.




(a) Use the graph to find an estimate for the median height of these plants.

.................................................... centimetres
**(1)**

(b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leqslant 40$

....................................................
**(1)**

(c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

....................................................
**(2)**

## Worked Solution 12

Read a cumulative-frequency graph

MODULAR UNIT 2
**4 marks**

### Median

$$\text{CF} = 25 \Rightarrow h \approx 28.$$

### Class frequency

$$\text{CF}(40) - \text{CF}(30) \approx 44 - 30 = 14.$$

### Above 35 cm

$$50 - \text{CF}(35) \approx 50 - 38 = 12.$$

### Final answer

$$28 \text{ cm}; \quad 14; \quad 12$$

## Original question 27

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks

**13** The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

| **Weight ($w$ kilograms)** | **Frequency** |
|---|---|
| $0 < w \leqslant 1$ | 4 |
| $1 < w \leqslant 2$ | 15 |
| $2 < w \leqslant 3$ | 20 |
| $3 < w \leqslant 4$ | 11 |
| $4 < w \leqslant 5$ | 6 |
| $5 < w \leqslant 6$ | 4 |

(a) Complete the cumulative frequency table.

| **Weight ($w$ kilograms)** | **Cumulative frequency** |
|---|---|
| $0 < w \leqslant 1$ | |
| $0 < w \leqslant 2$ | |
| $0 < w \leqslant 3$ | |
| $0 < w \leqslant 4$ | |
| $0 < w \leqslant 5$ | |
| $0 < w \leqslant 6$ | |

**(1)**

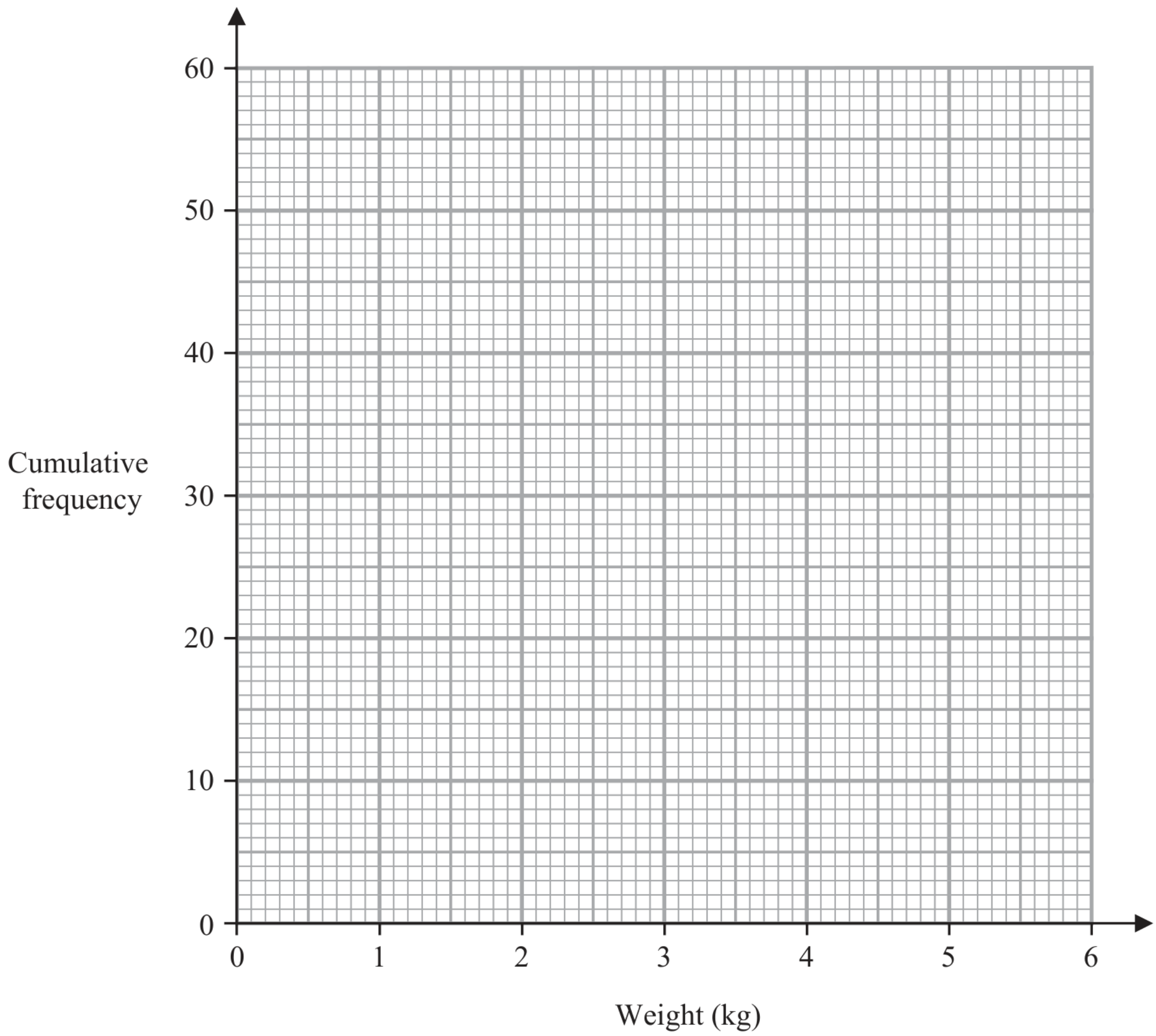
(b) On the grid opposite, draw a cumulative frequency graph for your table.

**(2)**

## Original question 27 (continued)

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2
**6 marks**




(c) Use your graph to find an estimate for the median weight of the 60 parcels.

........................................ kilograms
**(1)**

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

........................................
**(2)**

## Worked solution 27

MODULAR UNIT 2

Complete a cumulative-frequency table / Draw a cumulative-frequency graph / Read the median from the graph / Read the number above 3.7 kg

**6 marks**

### M01 – Complete a cumulative-frequency table

**Add frequencies successively**

$$4,\ 4+15=19,\ 19+20=39,\ 39+11=50,\ 50+6=56,\ 56+4=60.$$

**Final answer**

4, 19, 39, 50, 56, 60

### M02 – Draw a cumulative-frequency graph

**Plot upper-bound points**

$$(1,4),(2,19),(3,39),(4,50),(5,56),(6,60).$$

**Join**

Join the points with the accepted smooth curve or line segments.

**Final answer**

Correct cumulative-frequency graph

### M03 – Read the median from the graph

**Use the middle cumulative frequency**

Half of 60 is 30; reading across at CF=30 gives about 2.3–2.7 kg.

**Final answer**

2.3 to 2.7 kg

### M04 – Read the number above 3.7 kg

**Read CF at 3.7**

The graph gives approximately 45–48 parcels at or below 3.7 kg.

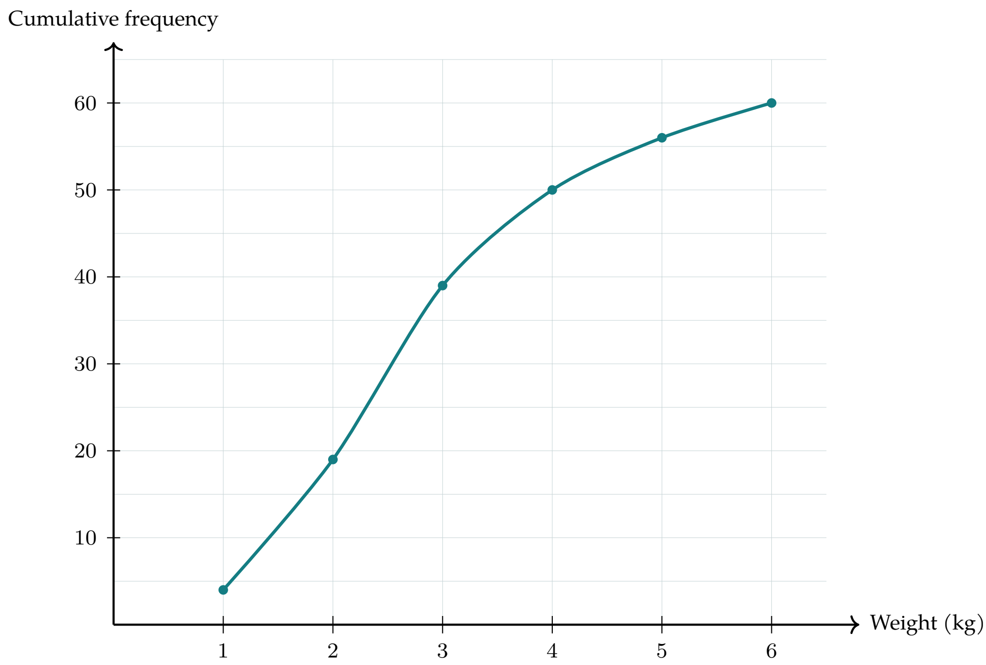
**Subtract from 60**

The number above 3.7 kg is 12–15.

**Final answer**

12 to 15 parcels

### Completed solution diagram: Completed cumulative-frequency graph



## Original question 26

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

ORIGINAL QUESTION
**7 marks**

**13** The table gives information about the ages, in years, of 80 people in a train carriage.

| **Age (*a* years)** | **Frequency** |
|---|---|
| $0 < a \leqslant 20$ | 7 |
| $20 < a \leqslant 30$ | 25 |
| $30 < a \leqslant 40$ | 20 |
| $40 < a \leqslant 50$ | 14 |
| $50 < a \leqslant 60$ | 8 |
| $60 < a \leqslant 70$ | 6 |

(a) Complete the cumulative frequency table.

| **Age (*a* years)** | **Cumulative frequency** |
|---|---|
| $0 < a \leqslant 20$ | |
| $0 < a \leqslant 30$ | |
| $0 < a \leqslant 40$ | |
| $0 < a \leqslant 50$ | |
| $0 < a \leqslant 60$ | |
| $0 < a \leqslant 70$ | |

**(1)**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

## Original question 26 (continued)

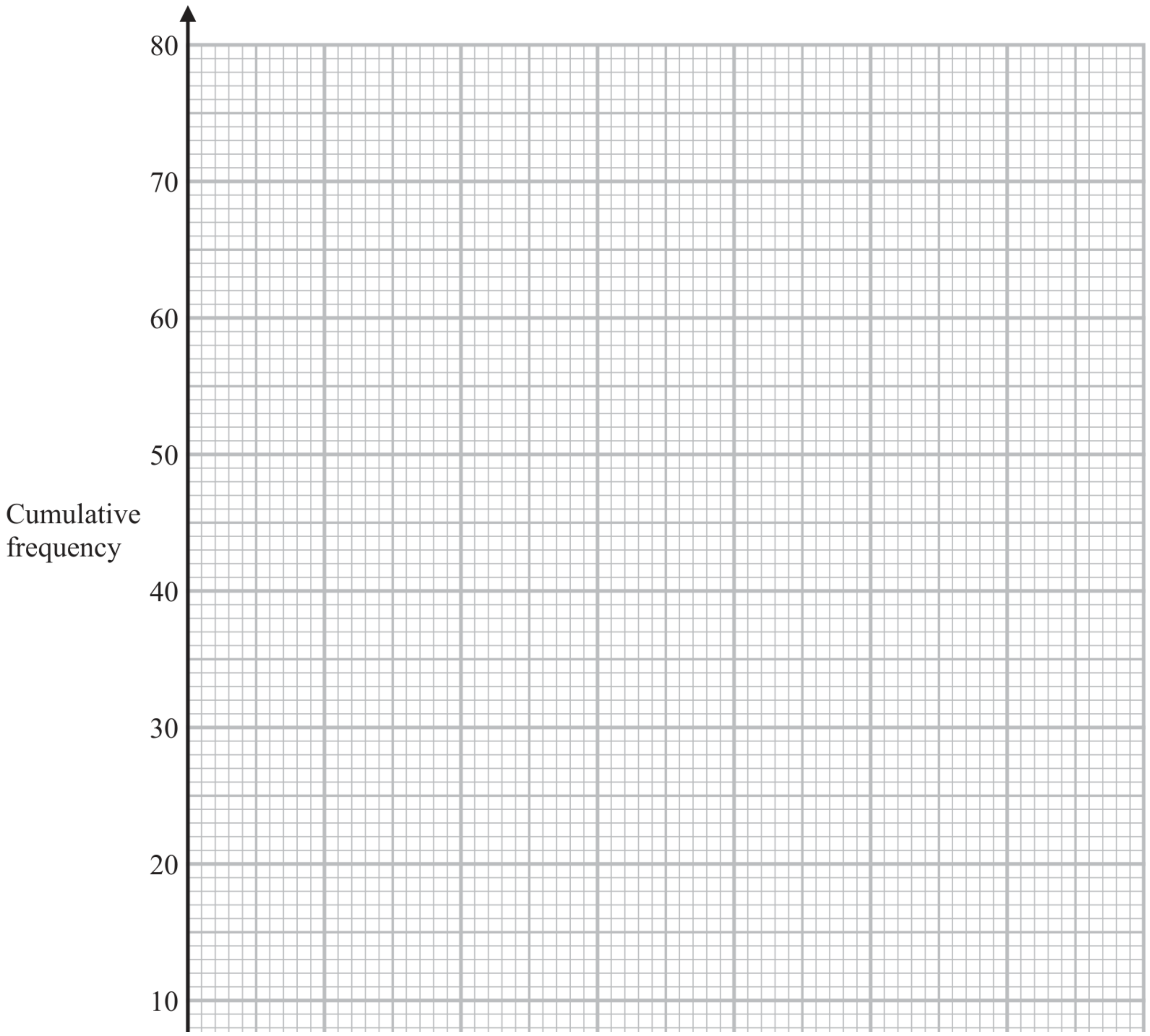
4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

ORIGINAL QUESTION
**7 marks**

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

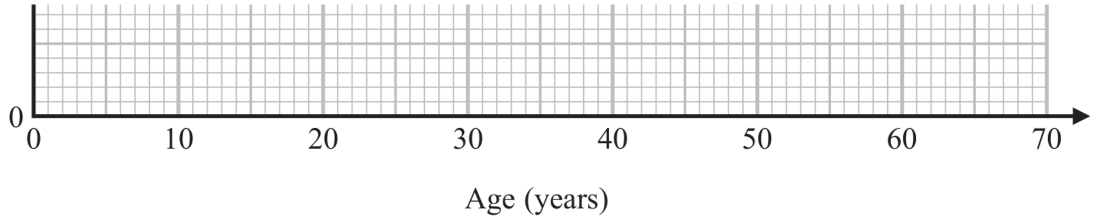
........................................ years
(1)



## Original question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

ORIGINAL QUESTION
**7 marks**




Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

(d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

.......................................................

(3)

## Worked solution 26

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2
7 marks

### Q13(a) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

**Method**
Successive cumulative totals are 7, 32, 52, 66, 74, 80.
*Why: Add each class frequency to the running total.*

**Final answer**
**cumulative frequencies:** 7, 32, 52, 66, 74, 80

### Q13(b) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Plot a cumulative-frequency curve from class frequencies

**Method**
Plot $(20, 7)$, $(30, 32)$, $(40, 52)$, $(50, 66)$, $(60, 74)$, and $(70, 80)$.
*Why: Cumulative frequencies are plotted at upper class boundaries.*

**Method**
Join the points with a smooth increasing curve or straight line segments.
*Why: A cumulative-frequency graph must preserve the cumulative order.*

**Final answer**
**graph:** CF points (20,7), (30,32), (40,52), (50,66), (60,74), (70,80)

### Q13(c) – Chapter: Cumulative Frequency Diagrams

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

**Method**
Read across from cumulative frequency 40 to the graph and then down to the age axis: the median is about 34 years.
*Why: The median is the 40th value out of 80.*

**Final answer**
**median age:** about 34 years

### Q13(d) – Chapter: Cumulative Frequency Diagrams

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

**Method**
From the graph, CF(18) is about 6 and CF(65) is about 77.
*Why: Use cumulative-frequency readings at both age limits.*

**Method**
About 77 - 6 = 71 people are aged between 18 and 65.
*Why: Subtract the lower cumulative count from the upper one.*

**Method**
60% of 71 is 42.6, so an estimate is 43 people.
*Why: Apply the percentage and give a whole-person estimate.*

**Final answer**
**people going to work:** about 43 people

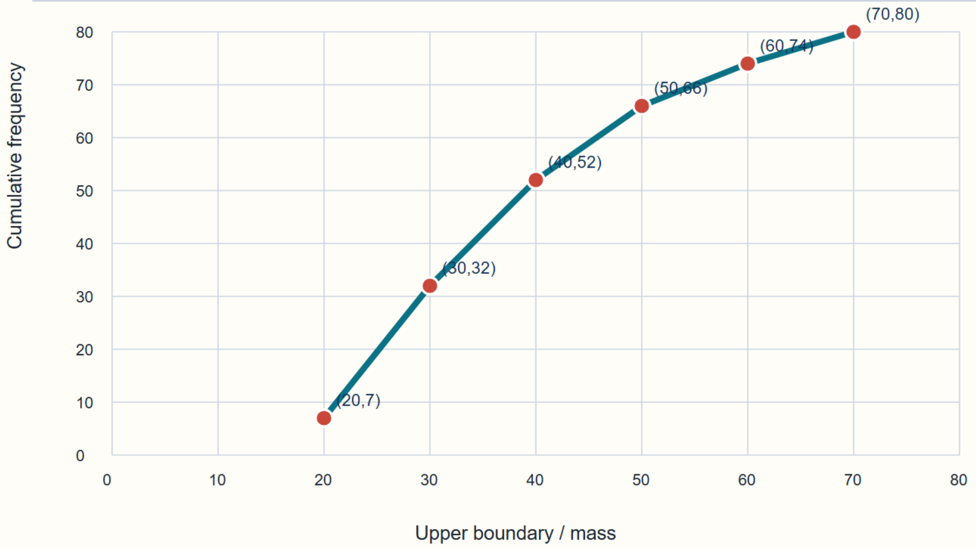
# Completed diagram 26

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2
**7 marks**

## Q13(b) Completed cumulative-frequency graph

Plot upper-bound points, then join in increasing order

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## Original question 09

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION
6 marks

**12** The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

| Time taken ($t$ minutes) | Frequency |
|---|---|
| $0 < t \leqslant 10$ | 7 |
| $10 < t \leqslant 20$ | 26 |
| $20 < t \leqslant 30$ | 24 |
| $30 < t \leqslant 40$ | 14 |
| $40 < t \leqslant 50$ | 7 |
| $50 < t \leqslant 60$ | 2 |

(a) Complete the cumulative frequency table.

| Time taken ($t$ minutes) | Cumulative frequency |
|---|---|
| $0 < t \leqslant 10$ | |
| $0 < t \leqslant 20$ | |
| $0 < t \leqslant 30$ | |
| $0 < t \leqslant 40$ | |
| $0 < t \leqslant 50$ | |
| $0 < t \leqslant 60$ | |

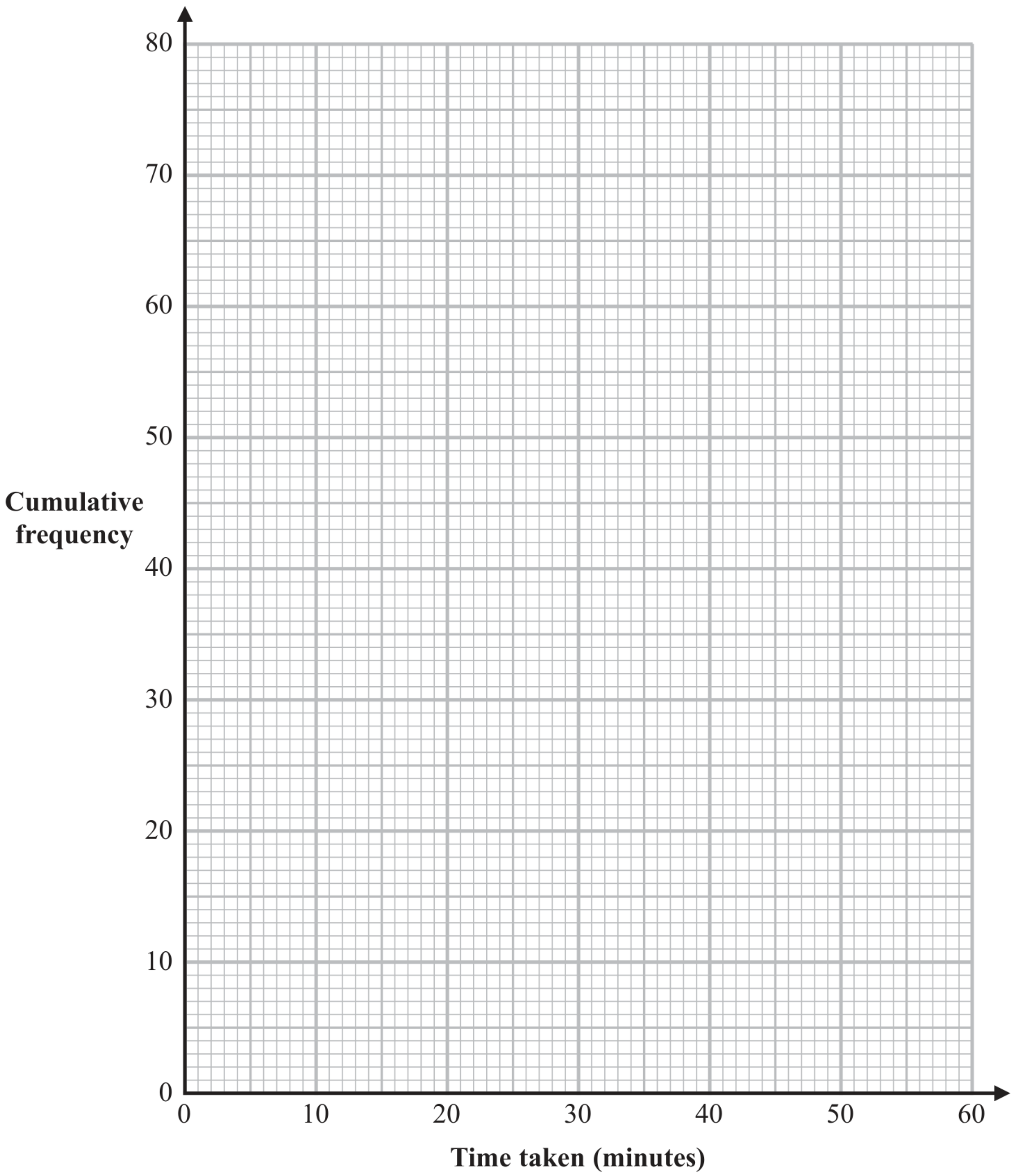
**(1)**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

## Original question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION
**6 marks**




(2)

## Original question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

ORIGINAL QUESTION
**6 marks**

(c) Use your graph to find an estimate for the median time taken.

.................................................... minutes
**(1)**

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

....................................................
**(2)**

## Worked solution 09

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2
**6 marks**

### Q12(a) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

**Method**
The cumulative frequencies are 7, 33, 57, 71, 78, 80.
*Why: Add the class frequencies successively.*

**Final answer**
**cumulative frequencies:** 7, 33, 57, 71, 78, 80

### Q12(b) – Chapter: Cumulative Frequency Diagrams

Family: Construct cumulative-frequency diagrams • Variant: Plot a cumulative-frequency curve from class frequencies

**Method**
Plot (10,7), (20,33), (30,57), (40,71), (50,78), (60,80).
*Why: Cumulative frequency is plotted at the upper class boundaries.*

**Method**
Join the points with straight or smooth connected segments.
*Why: A cumulative-frequency graph joins the plotted points in order.*

**Final answer**
**graph:** Correct cumulative-frequency curve through the six points

### Q12(c) – Chapter: Cumulative Frequency Diagrams

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

**Method**
Half of 80 is 40; reading the cumulative-frequency curve gives a median time in the range 21–24 minutes.
*Why: Use CF=40 and follow the reviewed graph from part (b).*

**Final answer**
**median:** 21–24 minutes

### Q12(d) – Chapter: Cumulative Frequency Diagrams

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate a percentage from a cumulative curve

**Method**

The graph gives about 72 customers with time at most 42 minutes.

*Why: Read the cumulative frequency at 42 minutes.*

**Method**

Customers taking more than 42 minutes $= 80 - 72 = 8$, so $P = 8/80 = 1/10$.

*Why: Subtract from the total and divide by 80.*

**Final answer**

**probability:** $8/80 = 1/10$

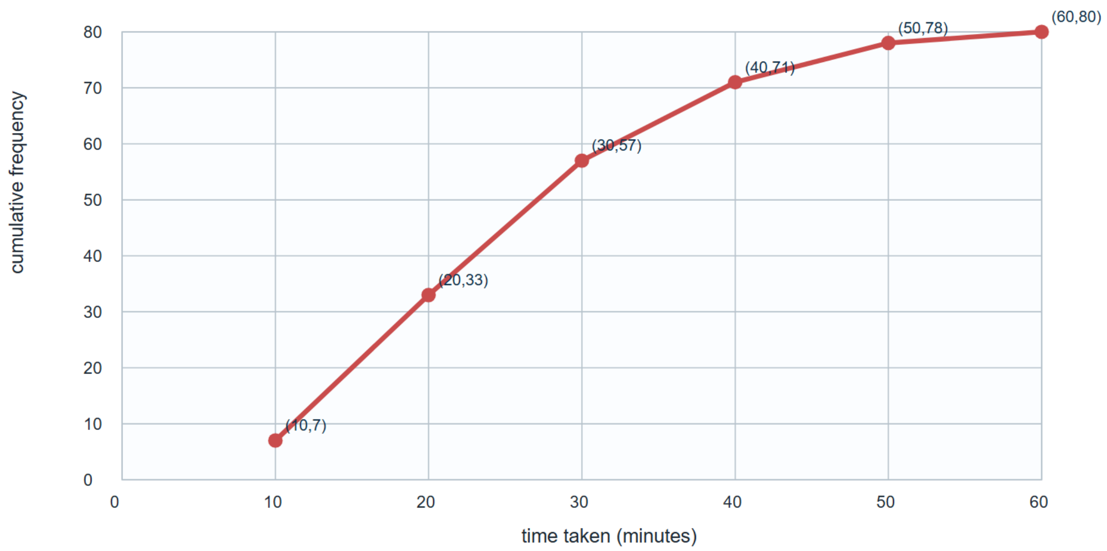
# Completed diagram 09

Chapter: Cumulative Frequency Diagrams

MODULAR UNIT 2
**6 marks**

## Q12 • Completed cumulative-frequency graph

Points at upper class boundaries

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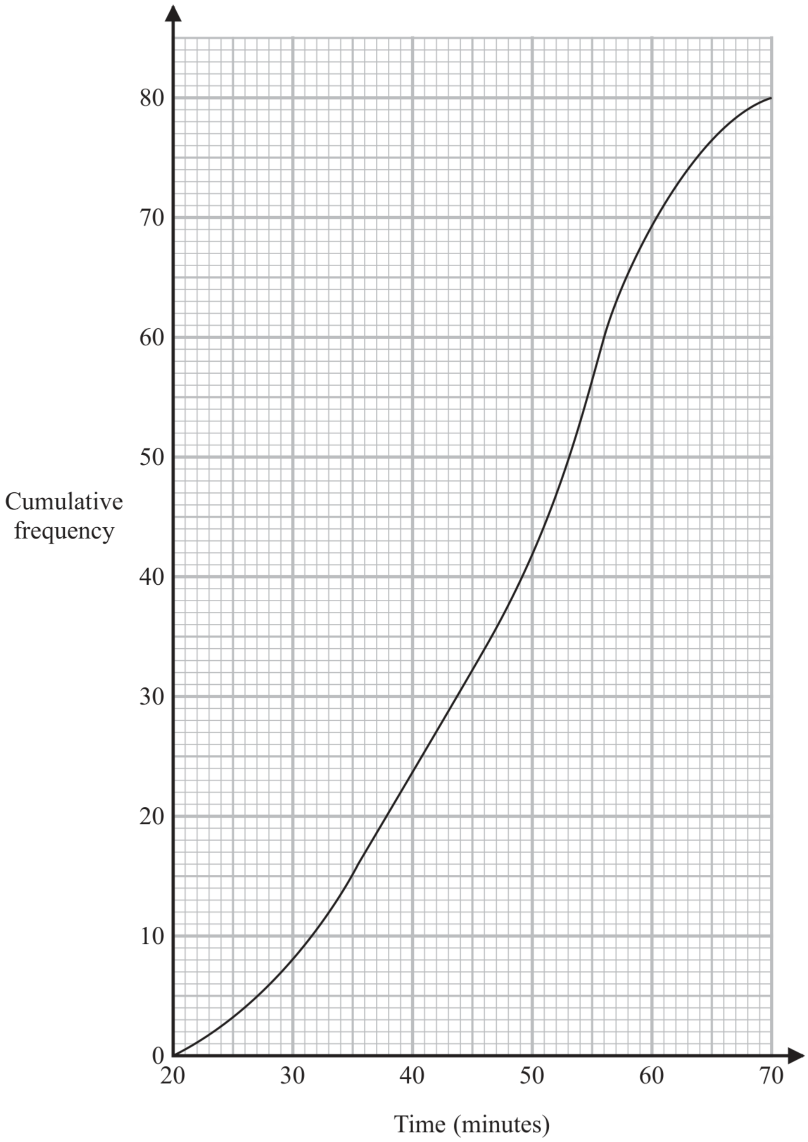
**12** A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.




(a) Use the graph to find an estimate for the interquartile range.

.............................................. minutes
**(2)**

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

(b) Work out an estimate for the number of men who took part in the race.

..............................................
**(3)**

## Worked solution

January 2020 | Question 12 | 2 marks

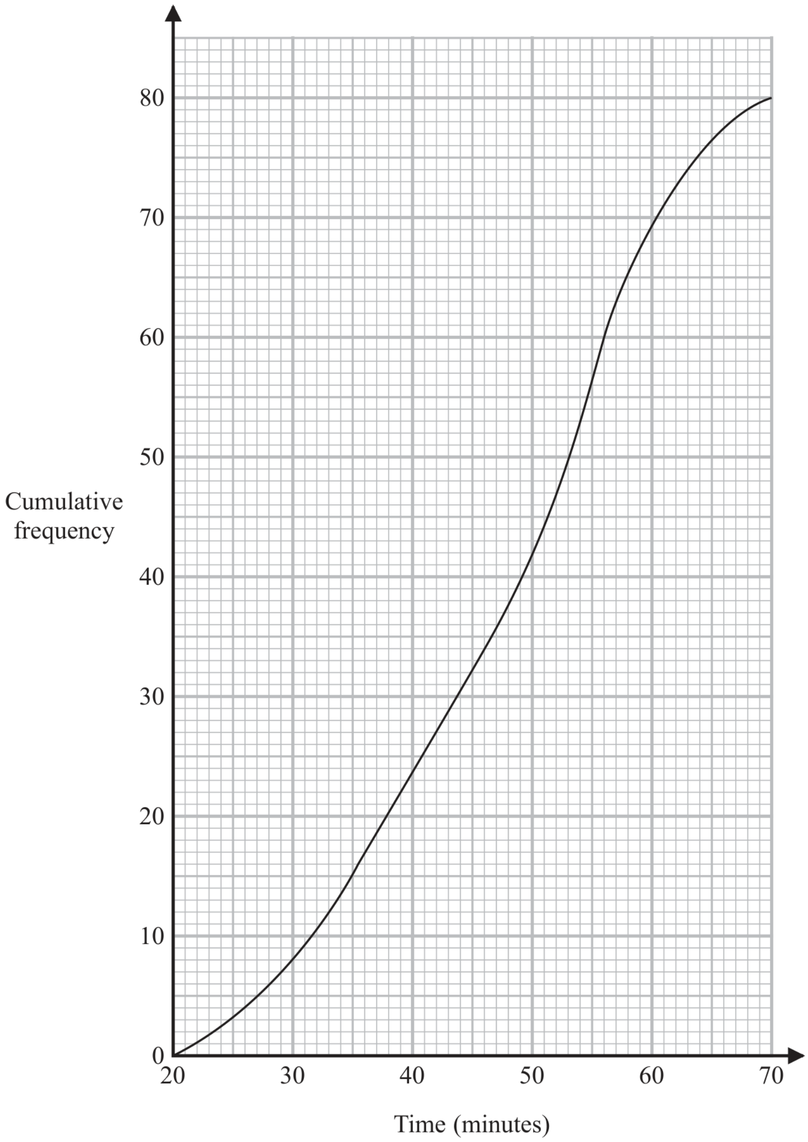
Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate IQR from a cumulative-frequency diagram

Method / working

1. Read Q1 at CF=20 (about 38 min).
2. Read Q3 at CF=60 (about 56 min), so IQR~56-38=18.

Final answer: about 18 minutes

**12** A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.




(a) Use the graph to find an estimate for the interquartile range.

.................................................. minutes

**(2)**

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

(b) Work out an estimate for the number of men who took part in the race.

..................................................

**(3)**

## Worked solution

January 2020 | Question 12 | 3 marks

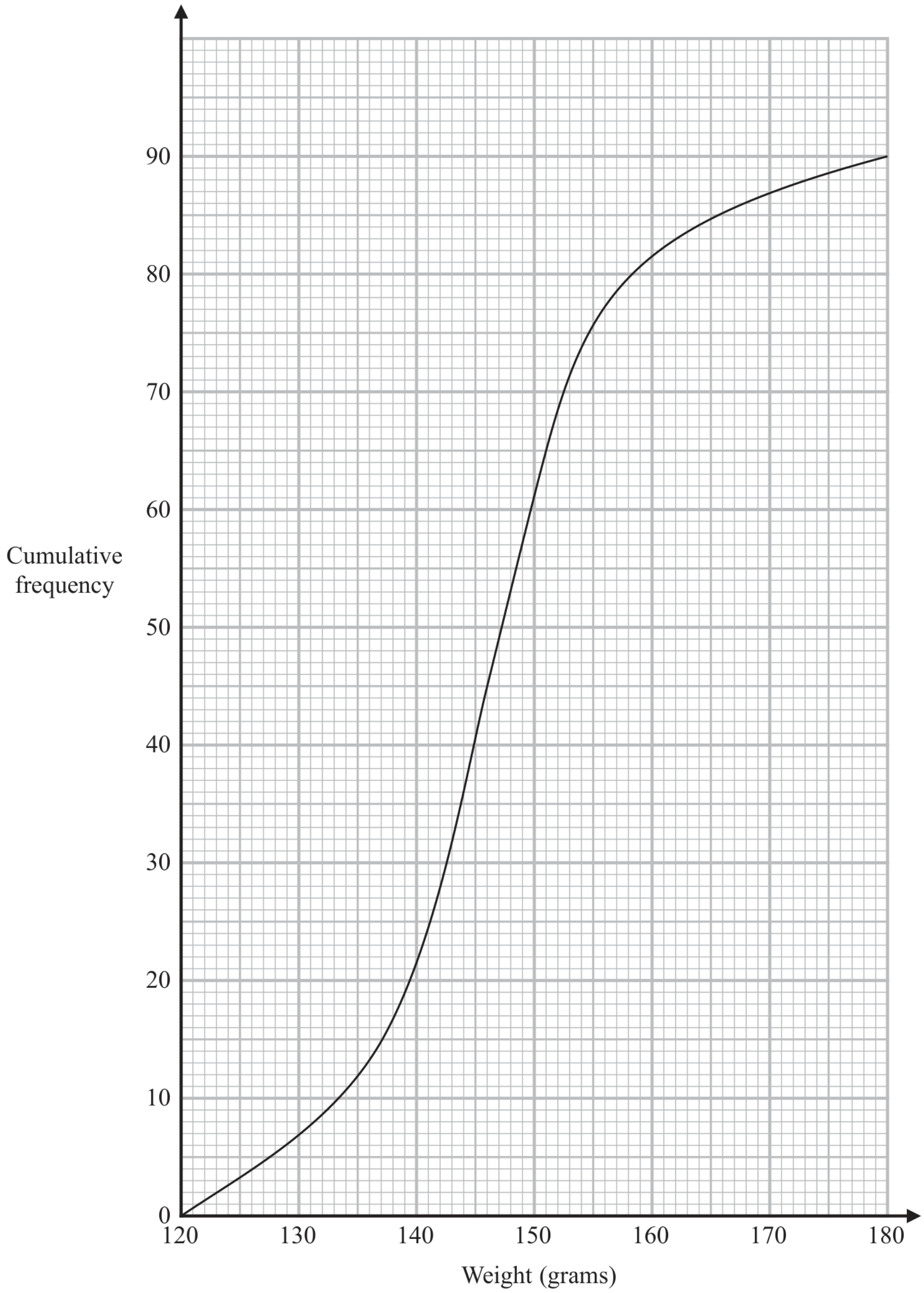
Family: Read counts and thresholds from cumulative frequency • Variant: Estimate a percentage from a cumulative curve

Method / working

1. At 50 minutes the graph gives about 42 people.
2. No women are at or below 50, so 42 is the number of men at or below 50.
3. 0.60 x men=42, hence men~70.

Final answer: about 70 men

**11** The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.




(a) Find an estimate for the median of the weights of these bags of sweets.

.................................................... grams

**(2)**

## Worked solution

January 2021 | Question 11 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method / working

1. Read the median at cumulative frequency 45.
2. The graph gives approximately 146 g (accepted range 145-147 g).

Final answer: 146 g

Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than $d$ grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

(b) Find the value of $d$.

$d$ = ..........................................................

**(3)**

## Worked solution

January 2021 | Question 11 | 3 marks

Family: Read counts and thresholds from cumulative frequency • Variant: Recover a threshold from a target count or amount

Method / working

1. Large-bag frequency = $93.75/3.75=25$.
2. So cumulative frequency at $d$ is $90-25=65$.
3. Reading $CF=65$ gives $d \sim 151$ g.

Final answer: $d=151$

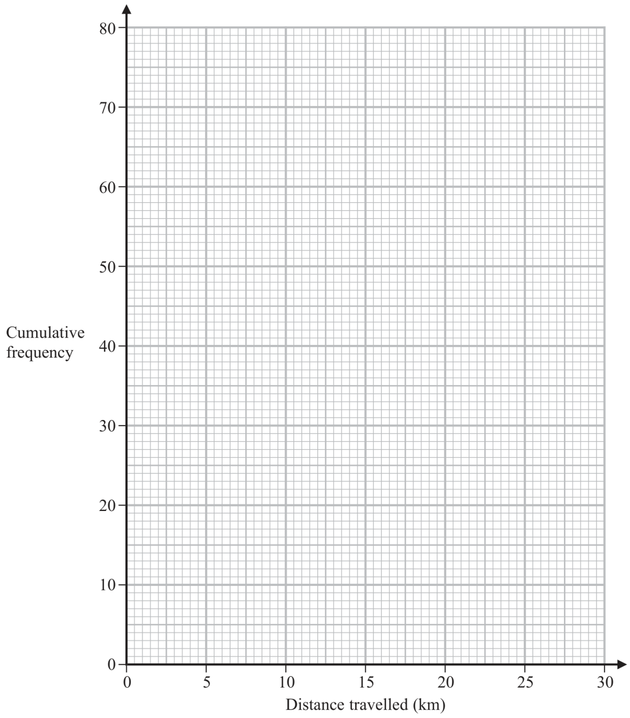
## June 2021 | 4MA1-2H | Question 12 | 2 marks

Chapter OL-T55: Cumulative Frequency Diagrams

**12** The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office $A$.

| Distance travelled ($d$ km) | Cumulative frequency |
|---|---|
| $0 < d \leqslant 5$ | 17 |
| $0 < d \leqslant 10$ | 32 |
| $0 < d \leqslant 15$ | 57 |
| $0 < d \leqslant 20$ | 70 |
| $0 < d \leqslant 25$ | 76 |
| $0 < d \leqslant 30$ | 80 |

(a) On the grid below, draw a cumulative frequency graph for the information in the table.




**(2)**

(b) Use your graph to find an estimate for the median distance travelled.

.................................................. km

**(1)**

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.................................................. km

**(2)**

For Office $B$, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office $A$ and workers at Office $B$ travel from home to work.
Write down **two** comparisons.

1 ..................................................

..................................................

..................................................

2 ..................................................

..................................................

..................................................

**(2)**

## Worked solution

June 2021 | Question 12 | 2 marks

Family: Construct cumulative-frequency diagrams • Variant: Plot from an existing cumulative-frequency table

Method / working

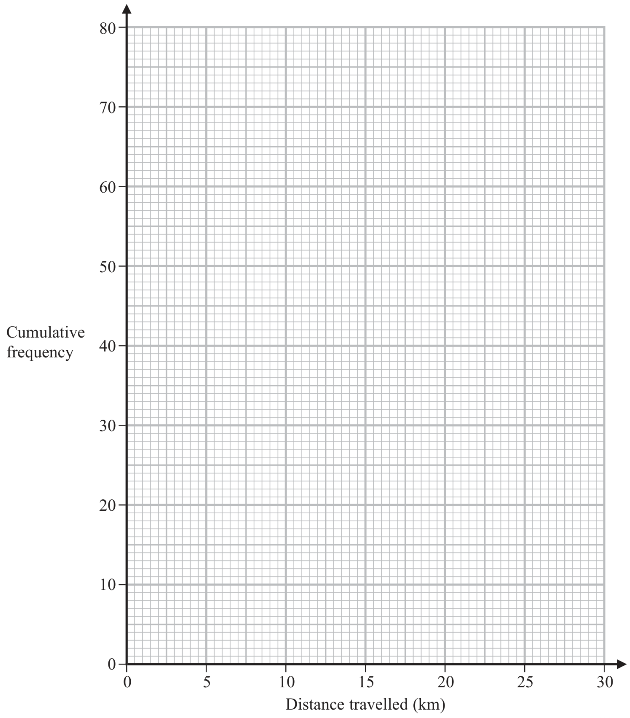
1. Plot the six cumulative-frequency points at the upper class boundaries (or accepted class midpoints).
2. Join them with a smooth increasing curve/line segments, not bars.

Final answer: Cumulative-frequency graph

**12** The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office $A$.

| **Distance travelled ($d$ km)** | **Cumulative frequency** |
|---|---|
| $0 < d \leqslant 5$ | 17 |
| $0 < d \leqslant 10$ | 32 |
| $0 < d \leqslant 15$ | 57 |
| $0 < d \leqslant 20$ | 70 |
| $0 < d \leqslant 25$ | 76 |
| $0 < d \leqslant 30$ | 80 |

(a) On the grid below, draw a cumulative frequency graph for the information in the table.




**(2)**

(b) Use your graph to find an estimate for the median distance travelled.

.......................................... km

**(1)**

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.......................................... km

**(2)**

For Office $B$, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office $A$ and workers at Office $B$ travel from home to work.
Write down **two** comparisons.

1 ..........................................................................................

..........................................................................................

..........................................................................................

2 ..........................................................................................

..........................................................................................

..........................................................................................

**(2)**

## Worked solution

June 2021 | Question 12 | 1 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate the median from a cumulative-frequency diagram

Method / working

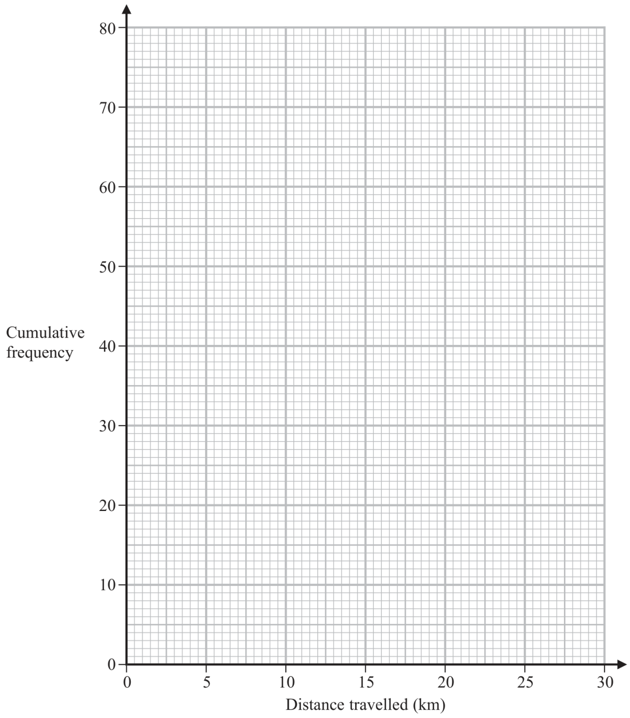
1. Read the graph at cumulative frequency 40 (the median position).
2. The estimate is in the accepted range 10.5-12 km.

Final answer: 10.5-12 km

**12** The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office *A*.

| **Distance travelled (*d* km)** | **Cumulative frequency** |
|---|---|
| $0 < d \leqslant 5$ | 17 |
| $0 < d \leqslant 10$ | 32 |
| $0 < d \leqslant 15$ | 57 |
| $0 < d \leqslant 20$ | 70 |
| $0 < d \leqslant 25$ | 76 |
| $0 < d \leqslant 30$ | 80 |

(a) On the grid below, draw a cumulative frequency graph for the information in the table.




**(2)**

(b) Use your graph to find an estimate for the median distance travelled.

.............................................. km

**(1)**

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.............................................. km

**(2)**

For Office *B*, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office *A* and workers at Office *B* travel from home to work.
Write down **two** comparisons.

1 ..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

2 ..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

**(2)**

## Worked solution

June 2021 | Question 12 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate IQR from a cumulative-frequency diagram

Method / working

1. Read lower and upper quartiles at CF 20 and CF 60.
2. Subtract the two readings to obtain an IQR in the accepted range 8.5-11.5 km.

Final answer: 8.5-11.5 km

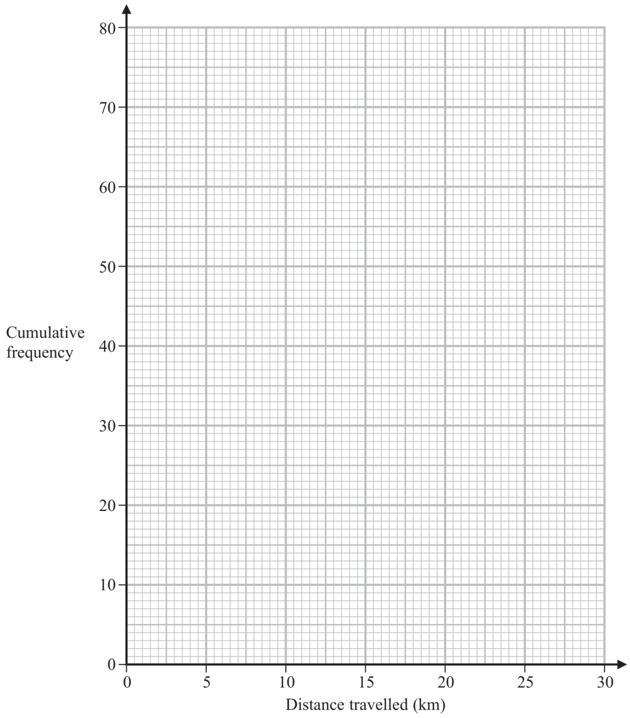
## June 2021 | 4MA1-2H | Question 12 | 2 marks

Chapter OL-T55: Cumulative Frequency Diagrams

**12** The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office *A*.

| Distance travelled ($d$ km) | Cumulative frequency |
|---|---|
| $0 < d \leqslant 5$ | 17 |
| $0 < d \leqslant 10$ | 32 |
| $0 < d \leqslant 15$ | 57 |
| $0 < d \leqslant 20$ | 70 |
| $0 < d \leqslant 25$ | 76 |
| $0 < d \leqslant 30$ | 80 |

(a) On the grid below, draw a cumulative frequency graph for the information in the table.




**(2)**

(b) Use your graph to find an estimate for the median distance travelled.

.............................................. km

**(1)**

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.............................................. km

**(2)**

For Office *B*, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office *A* and workers at Office *B* travel from home to work.
Write down **two** comparisons.

1 ..............................................................................................................

..............................................................................................................

..............................................................................................................

2 ..............................................................................................................

..............................................................................................................

..............................................................................................................

**(2)**

## Worked solution

June 2021 | Question 12 | 2 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Compare two distributions using cumulative-frequency measures

Method / working

1. Compare medians: Office B median 15 km is higher, so B workers tend to travel further.
2. Compare IQRs: Office A's IQR is larger than B's 5 km, so A distances are more varied.

Final answer: Office B travels further; Office A is more spread out

## November 2021 | 4MA1-1H | Question 11 | 1 marks

Chapter OL-T55: Cumulative Frequency Diagrams

**11** The table gives information about the times taken by 90 runners to complete a 10 km race.

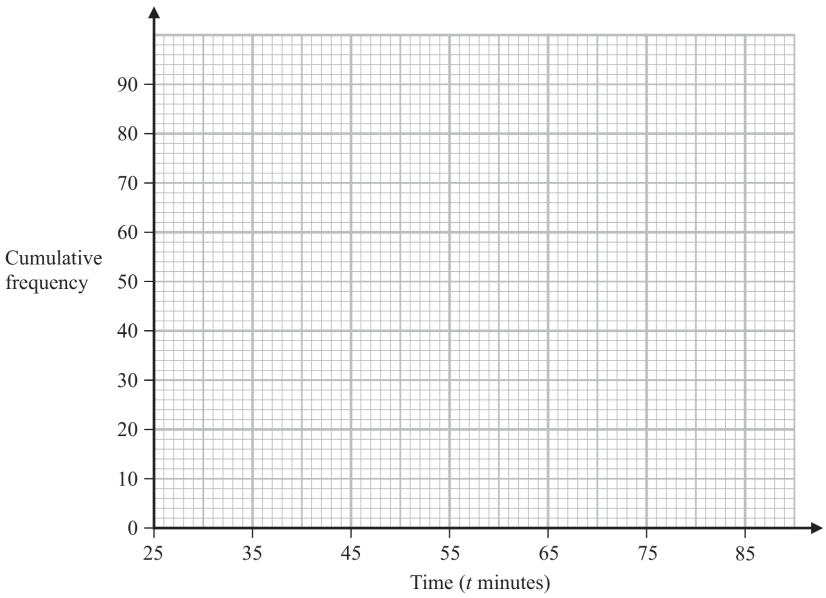
| Time ($t$ minutes) | Frequency |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $35 < t \leqslant 45$ | 24 |
| $45 < t \leqslant 55$ | 28 |
| $55 < t \leqslant 65$ | 12 |
| $65 < t \leqslant 75$ | 10 |
| $75 < t \leqslant 85$ | 4 |

(a) Complete the cumulative frequency table.

| Time ($t$ minutes) | Cumulative frequency |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $25 < t \leqslant 45$ | |
| $25 < t \leqslant 55$ | |
| $25 < t \leqslant 65$ | |
| $25 < t \leqslant 75$ | |
| $25 < t \leqslant 85$ | |

**(1)**

(b) On the grid below, draw a cumulative frequency graph for your table.




**(2)**

Any runner who completed the race in a time $T$ minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

........................................ runners

**(2)**

## Worked solution

November 2021 | Question 11 | 1 marks

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

Method / working
1. The cumulative frequencies are
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. 12,\ 36,\ 64,\ 76,\ 86,\ 90
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. For \(42<T\leq 52\),
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. F(52)\approx 36+frac{7}{10}(64-36)=55.6
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. F(42)\approx 12+frac{7}{10}(36-12)=28.8
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. 55.6-28.8=26.8
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. Answer: about \(27\) runners.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: about 27 runners.

**November 2021 | 4MA1-1H | Question 11 | 2 marks**

Chapter OL-T55: Cumulative Frequency Diagrams

**11** The table gives information about the times taken by 90 runners to complete a 10 km race.

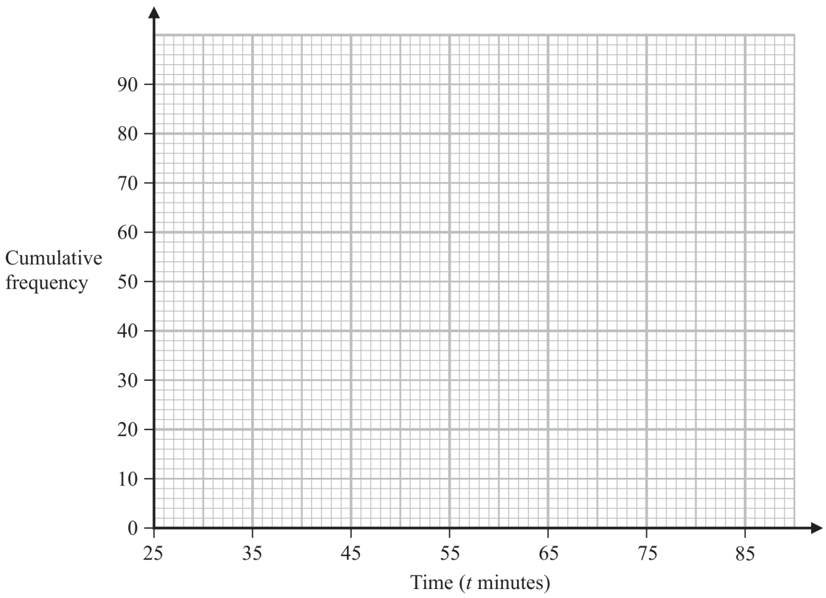
| **Time ($t$ minutes)** | **Frequency** |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $35 < t \leqslant 45$ | 24 |
| $45 < t \leqslant 55$ | 28 |
| $55 < t \leqslant 65$ | 12 |
| $65 < t \leqslant 75$ | 10 |
| $75 < t \leqslant 85$ | 4 |

(a) Complete the cumulative frequency table.

| **Time ($t$ minutes)** | **Cumulative frequency** |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $25 < t \leqslant 45$ | |
| $25 < t \leqslant 55$ | |
| $25 < t \leqslant 65$ | |
| $25 < t \leqslant 75$ | |
| $25 < t \leqslant 85$ | |

**(1)**

(b) On the grid below, draw a cumulative frequency graph for your table.




**(2)**

Any runner who completed the race in a time $T$ minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.............................................. runners

**(2)**

## Worked solution

November 2021 | Question 11 | 2 marks

Family: Construct cumulative-frequency diagrams • Variant: Plot a cumulative-frequency curve from class frequencies

Method / working
1. The cumulative frequencies are
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $12,\ 36,\ 64,\ 76,\ 86,\ 90$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. For $42<T\leq 52$,
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $F(52)\approx 36+\frac{7}{10}(64-36)=55.6$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. $F(42)\approx 12+\frac{7}{10}(36-12)=28.8$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. $55.6-28.8=26.8$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. Answer: about $27$ runners.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: about 27 runners.

**11** The table gives information about the times taken by 90 runners to complete a 10 km race.

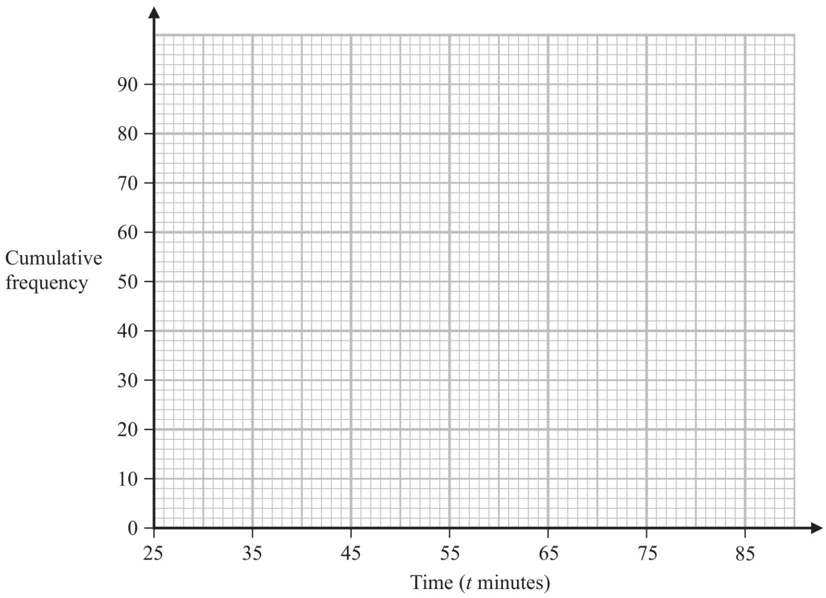
| Time ($t$ minutes) | Frequency |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $35 < t \leqslant 45$ | 24 |
| $45 < t \leqslant 55$ | 28 |
| $55 < t \leqslant 65$ | 12 |
| $65 < t \leqslant 75$ | 10 |
| $75 < t \leqslant 85$ | 4 |

(a) Complete the cumulative frequency table.

| Time ($t$ minutes) | Cumulative frequency |
|---|---|
| $25 < t \leqslant 35$ | 12 |
| $25 < t \leqslant 45$ | |
| $25 < t \leqslant 55$ | |
| $25 < t \leqslant 65$ | |
| $25 < t \leqslant 75$ | |
| $25 < t \leqslant 85$ | |

**(1)**

(b) On the grid below, draw a cumulative frequency graph for your table.




**(2)**

Any runner who completed the race in a time $T$ minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.................................................. runners

**(2)**

## Worked solution

November 2021 | Question 11 | 2 marks

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

Method / working

1. The cumulative frequencies are

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $12,\ 36,\ 64,\ 76,\ 86,\ 90$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. For $42<T\leq 52$,

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $F(52)\approx 36+\frac{7}{10}(64-36)=55.6$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $F(42)\approx 12+\frac{7}{10}(36-12)=28.8$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $55.6-28.8=26.8$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Answer: about $27$ runners.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: about 27 runners.

**10** The table shows information about the number of minutes each of 120 buses was late last Monday.

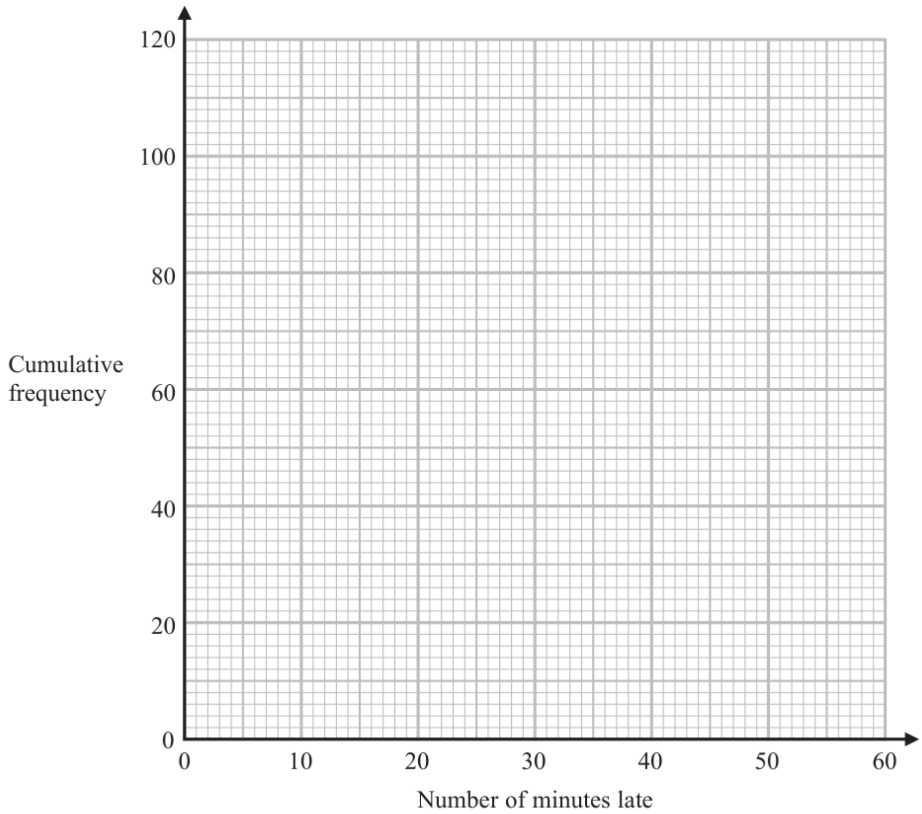
| **Number of minutes late ($L$)** | **Frequency** |
|---|---|
| $0 < L \leqslant 10$ | 10 |
| $10 < L \leqslant 20$ | 16 |
| $20 < L \leqslant 30$ | 44 |
| $30 < L \leqslant 40$ | 29 |
| $40 < L \leqslant 50$ | 15 |
| $50 < L \leqslant 60$ | 6 |

(a) Complete the cumulative frequency table below.

| **Number of minutes late ($L$)** | **Cumulative frequency** |
|---|---|
| $0 < L \leqslant 10$ | |
| $0 < L \leqslant 20$ | |
| $0 < L \leqslant 30$ | |
| $0 < L \leqslant 40$ | |
| $0 < L \leqslant 50$ | |
| $0 < L \leqslant 60$ | |

**(1)**

(b) On the grid, draw a cumulative frequency graph for your table.




**(2)**

(c) Use your graph to find an estimate for the interquartile range.

.......................................minutes

**(2)**

(d) Use your graph to find an estimate for the number of buses that were more than 48 minutes late last Monday.

.......................................

**(2)**

**(Total for Question 10 is 7 marks)**

## Answer reference | May-Nov 2020 | Question 10

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

| | | | | | | |
|---|---|---|---|---|---|---|
| **10** | a | | 10, 26, 70, 99, 114, 120 | 1 | B1 | |
| | b | | correct cumulative frequency graph | 2 | B2 | fully correct cf graph – points at ends of intervals and joined with curve or line segments<br><br>If not B2 then B1<br>for 5 or 6 (ft from a table with only one arithmetic error) of their points at ends of intervals and joined with curve or line segments<br><br>**OR** for 5 or 6 points plotted correctly at ends of intervals not joined<br><br>**OR** for 5 or 6 of their points from table plotted consistently within each interval (not at upper ends of intervals) at their correct heights and joined with smooth curve or line segments |
| | c | | | | M1 | For use of 30 and 90, or 30.25 and 90.75 (eg reading of 21 and 37 stated or indicated by marks on horizontal axis that correspond to 30 (or 30.25) and 90 (or 90.75) on the vertical axis or correct readings ft their cf graph provided method to show readings is shown) |
| | | | 16 | 2 | A1 | accept 14 – 18, ft from their cf graph (ft provided method to show readings is shown) |
| | d | | | | M1 | For use of cf from number of minutes late being 48 (eg an indication by a mark on the vertical axis corresponding to 48 mins late or a correct reading ft their cf graph) |
| | | | 9 | 2 | A1 | accept 7 – 10, ft from their cf graph |
| | | | | | | **Total 7 marks** |

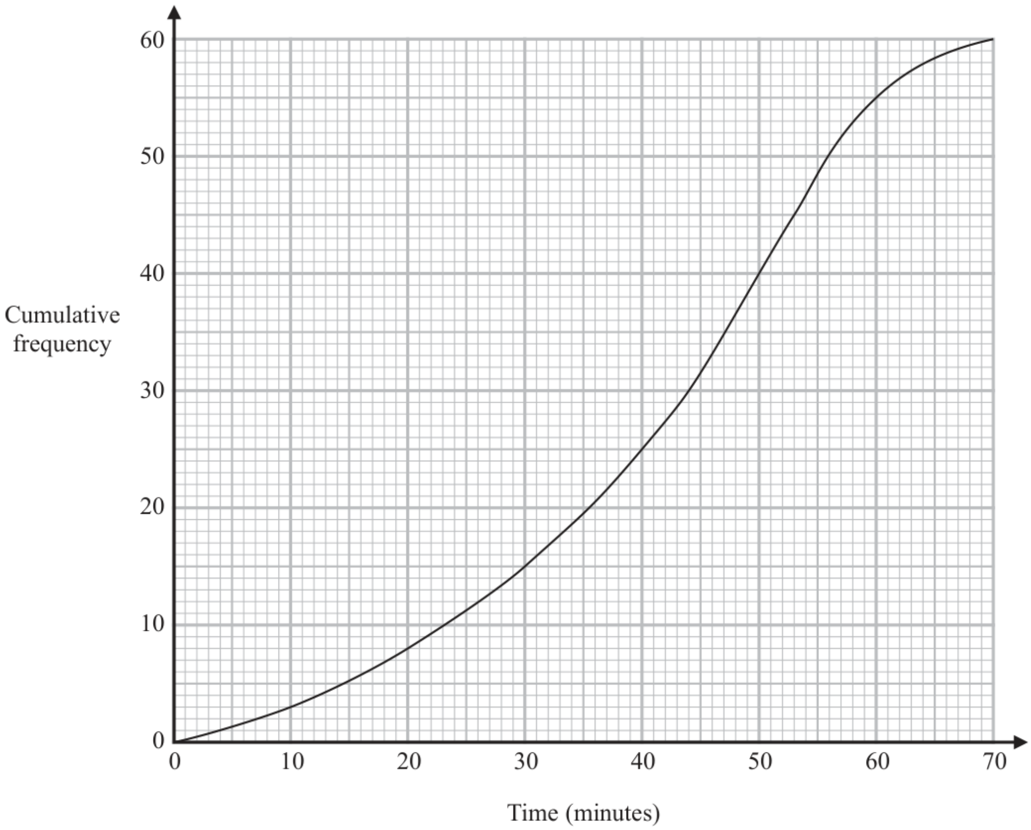
## Worked solution

May-Nov 2020 | Question 10 | 7 marks

Family: Construct cumulative-frequency diagrams • Variant: Complete cumulative totals from a frequency table

Official final mark-scheme pages are embedded immediately after this question.

**12** The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.




(a) Use the graph to find an estimate for the median time people took to shop in the market.

.......................................... minutes

**(1)**

(b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

..........................................

**(2)**

(c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

| **Time taken to shop in the market ($m$ minutes)** | **Frequency** |
|---|---|
| $0 < m \leqslant 10$ | 3 |
| $10 < m \leqslant 20$ | 5 |
| $20 < m \leqslant 30$ | |
| $30 < m \leqslant 40$ | |
| $40 < m \leqslant 50$ | |
| $50 < m \leqslant 60$ | |
| $60 < m \leqslant 70$ | |

**(2)**

**(Total for Question 12 is 5 marks)**

## Answer reference | January 2023 | Question 12

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

<table>
<tr><th>Question</th><th></th><th>Working</th><th>Answer</th><th>Mark</th><th></th><th>Notes</th></tr>
<tr><td>12</td><td>(a)</td><td></td><td>43.5 - 44.5</td><td>1</td><td>B1</td><td>±0.5 small square</td></tr>
<tr><td></td><td>(b)</td><td>eg reading of 48 - 49</td><td></td><td>2</td><td>M1</td><td>For correct method to start the question eg a vertical line from 55 up to the line and a horizontal line from the correct point on the curve or a mark on the curve at the correct point and a mark on the vertical axis at the correct point<br>or a correct reading of 48 to 49</td></tr>
<tr><td></td><td></td><td>Correct answer scores full marks (unless from obvious incorrect working)</td><td>11 or 12</td><td></td><td>A1</td><td>Allow an answer of 11 or 12 (ie must be whole number)</td></tr>
<tr><td></td><td>(c)</td><td>
<table>
<tr><th>Time taken to shop in the market (m minutes)</th><th>Frequency</th></tr>
<tr><td>$0 < m \leq 10$</td><td>3</td></tr>
<tr><td>$10 < m \leq 20$</td><td>5</td></tr>
<tr><td>$20 < m \leq 30$</td><td>7</td></tr>
<tr><td>$30 < m \leq 40$</td><td>10</td></tr>
<tr><td>$40 < m \leq 50$</td><td>15</td></tr>
<tr><td>$50 < m \leq 60$</td><td>15</td></tr>
<tr><td>$60 < m \leq 70$</td><td>5</td></tr>
</table>
</td><td></td><td>2</td><td>B2</td><td>All values correctly filled in (NB: first 2 are already completed)<br>(B1 for 3 or 4 correct values from 7, 10, 15, 15, 5)</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td>Total 5 marks</td></tr>
</table>

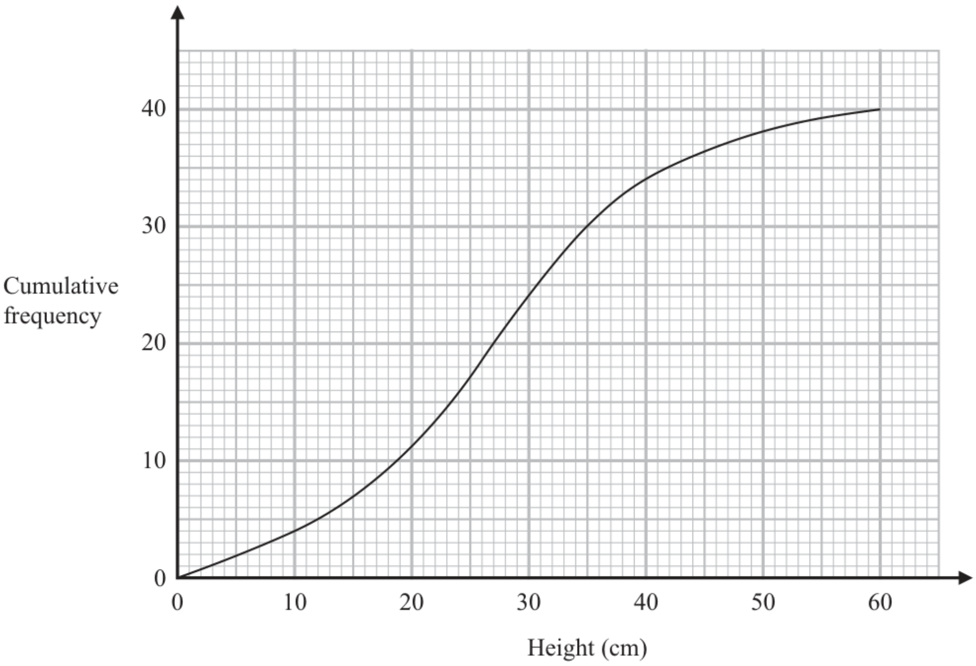
## Worked solution

January 2023 | Question 12 | 5 marks

Family: Read counts and thresholds from cumulative frequency • Variant: Estimate the number above or between thresholds

Official final mark-scheme pages are embedded immediately after this question.

**14** The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.




(a) Use the graph to find an estimate for the median height.

.................................................... cm

**(1)**

(b) Use the graph to find an estimate for the interquartile range of the heights.

.................................................... cm

**(2)**

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

........................................................ euros

**(2)**

**(Total for Question 14 is 5 marks)**

## Worked solution

November 2025 | Question 14 | 5 marks

Family: Estimate and compare location and spread from cumulative frequency • Variant: Estimate lower and upper quartiles from cumulative frequency

Solution
There are 40 plants.
From the graph,
medianapprox 26 cm
Q_1approx 19 cm,qquad Q_3approx 35 cm
IQRapprox 35-19=16 cm
For premium plants, use height >40 cm.
F(40)approx 34
40-34=6
6 x 30=180
Answers: median about 26 cm, IQR about 16 cm, and 180 euros.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION